

EXPERIMENT 6

IMPLEMENTING FABRIC CHANNELS FOR PRIVACY

Aim

To create and configure multiple channels in a Hyperledger Fabric network, deploy chaincode on a specific channel, and demonstrate how Fabric channels provide data privacy and isolation between organizations.

Software Requirements

Kali Linux / Ubuntu

Docker

Docker Compose

Node.js (Version 18 or later)

npm

Git

Hyperledger Fabric Samples

Hyperledger Fabric Binaries

Hyperledger Fabric Docker Images

Visual Studio Code (Optional)

Theory

A Fabric Channel is a private communication mechanism that allows a subset of organizations to maintain a separate ledger and execute transactions confidentially. Each channel has its own blockchain ledger, smart contracts (chaincode), and world state database. Organizations that are not members of a channel cannot access its ledger or transaction data. Channels enable enterprise applications to maintain confidentiality while sharing the same Hyperledger Fabric network.

Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

```
dhmarshna@DHARSHU:~$ cd ~/fabric-dev/fabric-samples/test-network
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

```
peerjsn - type common-envelope - output /home/dhmarshna/fabric-dev/fabric-samples/test-network/channel-2-artifacts/org2ns1anchor
2026-09-08 16:04:57.994 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-09-08 16:04:58.009 UTC 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ S|
```

Step 3: Verify the Existing Channel

peer channel list

```
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer channel list
2026-09-08 16:07:15.871 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Channels peers has joined:
mychannel
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 4: Create a New Private Channel

./network.sh createChannel -c privatechannel

```
pe.json --type common.Envelope --output /home/dhmarshna/fabric-dev/fabric-samples/test-network/channel-artifacts/Org2MSPanchors.tx
2026-09-08 16:07:59.654 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-09-08 16:07:59.677 UTC 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'privatechannel'
Channel 'privatechannel' joined
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 5: Verify Available Channels

peer channel list

```
2026-09-08 16:08:38.607 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Channels peers has joined:
mychannel
privatechannel
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 6: Deploy Chaincode on the Private Channel

./network.sh deployCC \
-c privatechannel \
-ccn basic \
-ccl javascript \
-ccp ../asset-transfer-basic/chaincode-javascript

```
+ peer lifecycle chaincode querycommitted --channelID privatechannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'privatechannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vsc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'privatechannel'
Chaincode initialization is not required
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 7: Initialize the Ledger

peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile \$ORDERER_CA \
-C privatechannel \
-n basic \
-c '{"function": "InitLedger", "Args": []}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles \$PEER0_ORG1_CA \

```
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $PEER0_ORG2_CA
```

```
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $ORDERER_CA \  
-C privatechannel \  
-n basic \  
-c '{"function":"InitLedger","Args":[]}' \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $PEER0_ORG1_CA \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $PEER0_ORG2_CA  
2026-09-08 16:12:36.286 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200  
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 8: Query Assets from the Private Channel

```
peer chaincode query \  
-C privatechannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'
```

```
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \  
-C privatechannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'  
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red",  
"ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size"  
:10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedVal  
ue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asse  
t6","Owner":"Michel","Size":15,"docType":"asset"}]  
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 9: Verify Channel Isolation

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset1"]}'
```

```
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset1"]}'  
Error: endorsement failure during query. response: status:500 message:"make sure the chaincode basic has been successfully defined on  
channel mychannel and try again: chaincode basic not found"  
dhmarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 10: Stop the Fabric Network

```
./network.sh down
```

```
dhارشنا@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 10/10
✓Container ca_org1 Removed 10.6s
✓Container ca_orderer Removed 10.9s
✓Container orderer.example.com Removed 0.6s
✓Container ca_org2 Removed 10.7s
✓Container peer0.org2.example.com Removed 1.2s
✓Container peer0.org1.example.com Removed 1.4s
✓Volume compose_peer0.org2.example.com Removed 0.1s
✓Network fabric_test Removed 0.4s
✓Volume compose_peer0.org1.example.com Removed 0.0s
✓Volume compose_orderer.example.com Removed 0.0s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic.1.0-1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc14976
15f5e33a8bfa908e0fd3555f864cfa1398b52d341cb:latest
Deleted: sha256:5e969d5e4d536ffcac2a7b26f7871abbbc605e64fd0850c75ff940a2935c1b84
Untagged: dev-peer0.org1.example.com-basic.1.0-1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433a
c0e74bfaeade1cf0b97f5c1163835f604aa885d60dc:latest
Deleted: sha256:e7bcf4fbfalaee1e15f3d50bedba30de605a5e5a249dcf73d5c3d99ad73b5983
dhارشنا@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Expected Key Observations

- peer channel list initially shows mychannel.
- privatechannel is created successfully.
- peer channel list shows mychannel and privatechannel.
- basic chaincode is deployed on privatechannel.
- Ledger initialization on privatechannel completes successfully.
- GetAllAssets retrieves assets from privatechannel.
- The separate channel demonstrates ledger/data isolation.
- The Fabric network is stopped successfully.

Result

Multiple Fabric channels were created and configured successfully. Chaincode was deployed on the private channel, ledger data was initialized and queried, and channel isolation was demonstrated, showing how Fabric channels provide privacy and data separation.